

# Functional PCA

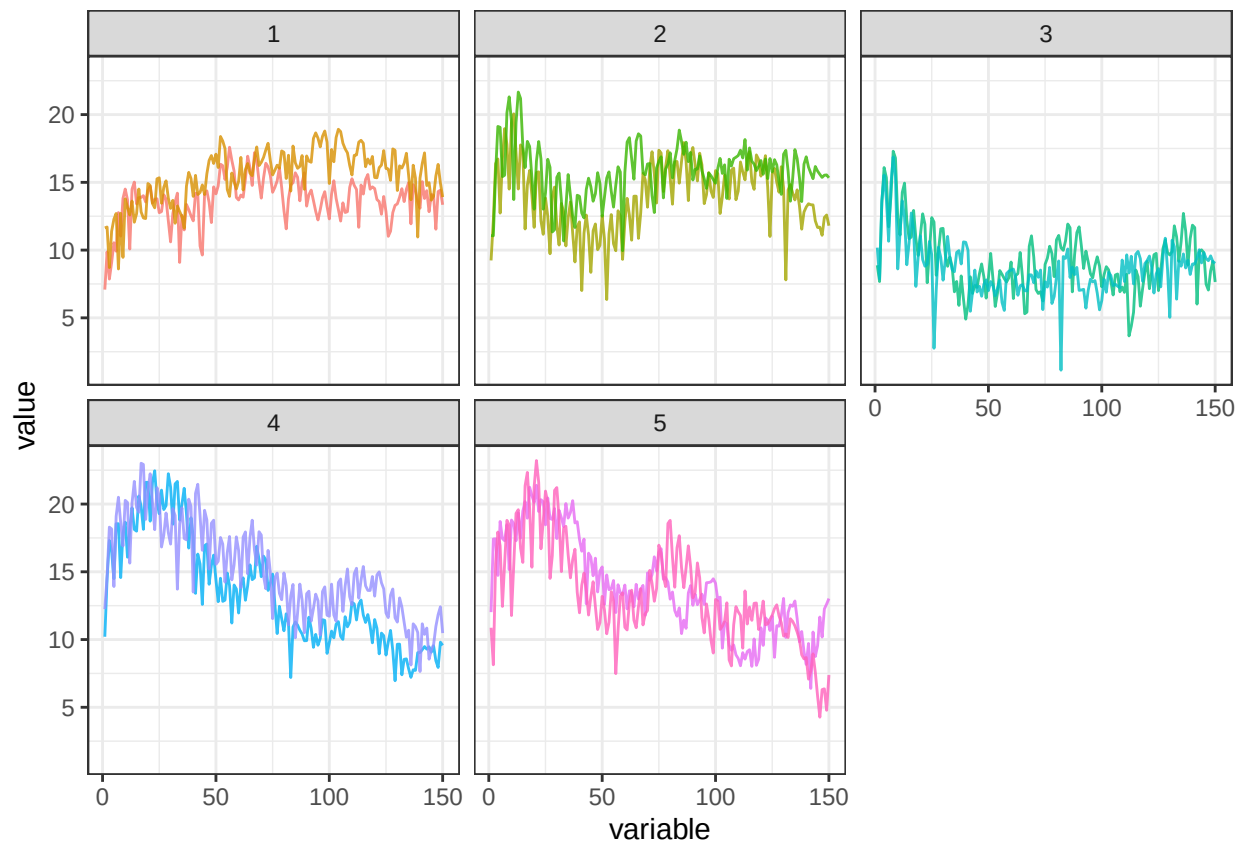
## Datos reales

```
data(phoneme)
mlearn<-phoneme[["learn"]]
glearn<-phoneme[["classlearn"]]
mtest<-phoneme[["test"]]
gtest<-phoneme[["classtest"]]
dataf<-data.frame(glearn)

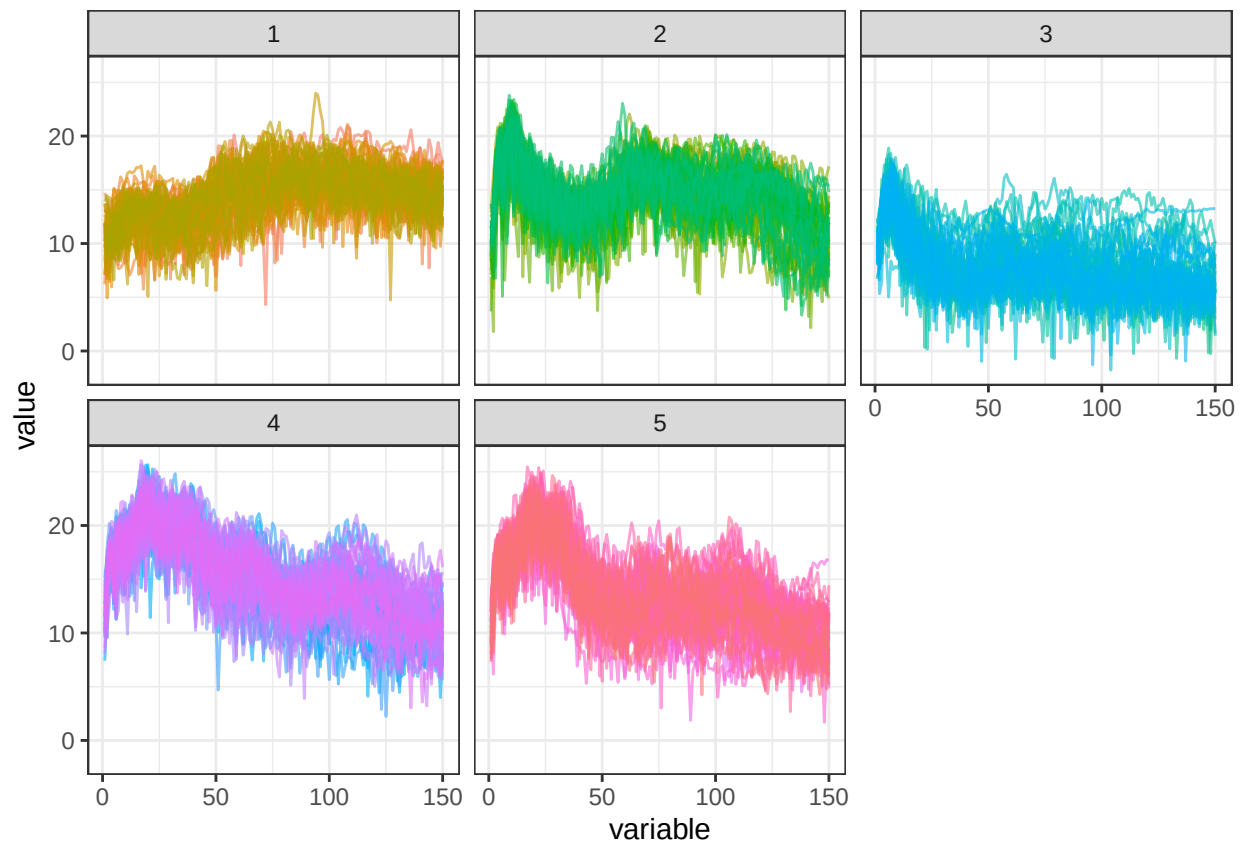
set.seed(1234)
ids <- sample(nrow(mlearn$data), 50)

datos <- mlearn$data %>%
  as.data.table %>%
  .[, y := glearn] %>%
  .[, id := 1:nrow(.)] %>%
  melt(id = c('y', 'id')) %>%
  .[, variable := as.numeric(gsub('X', '', variable))]

datos %>%
  .[(id %% 25) == 0] %>%
  # .[id %in% ids] %>%
  # .[y != 5] %>%
  ggplot(aes(variable, value, col = factor(id), group = id)) +
  geom_line(alpha = .8, show.legend = F) +
  facet_wrap(~y) +
  theme_bw()
```



```
datos %>%
  ggplot(aes(variable, value, col = factor(id), group = id)) +
  geom_line(alpha = .6, show.legend = F) +
  facet_wrap(~y) +
  theme_bw()
```



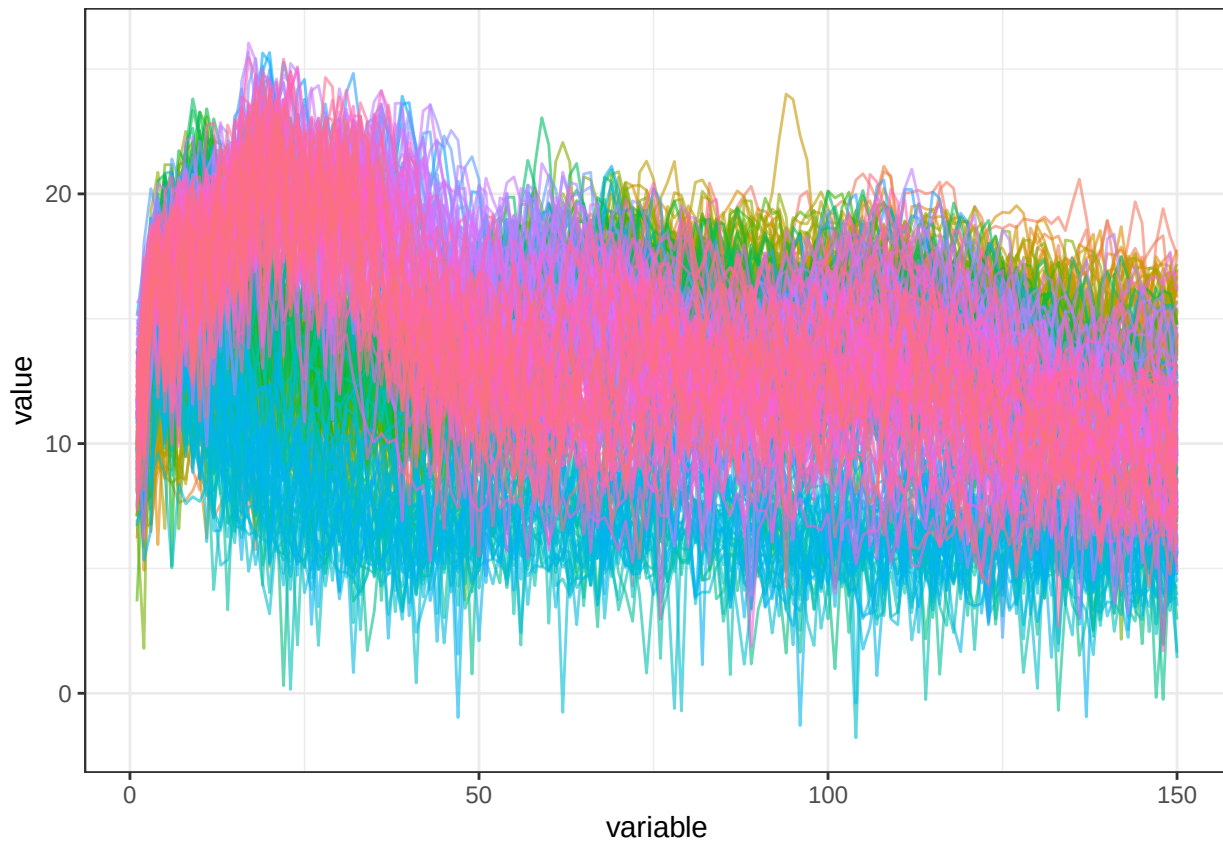
```
fd <- Data2fd(mlearn$data %>% t)
```

```
## 'y' is missing, using 'argvals'
```

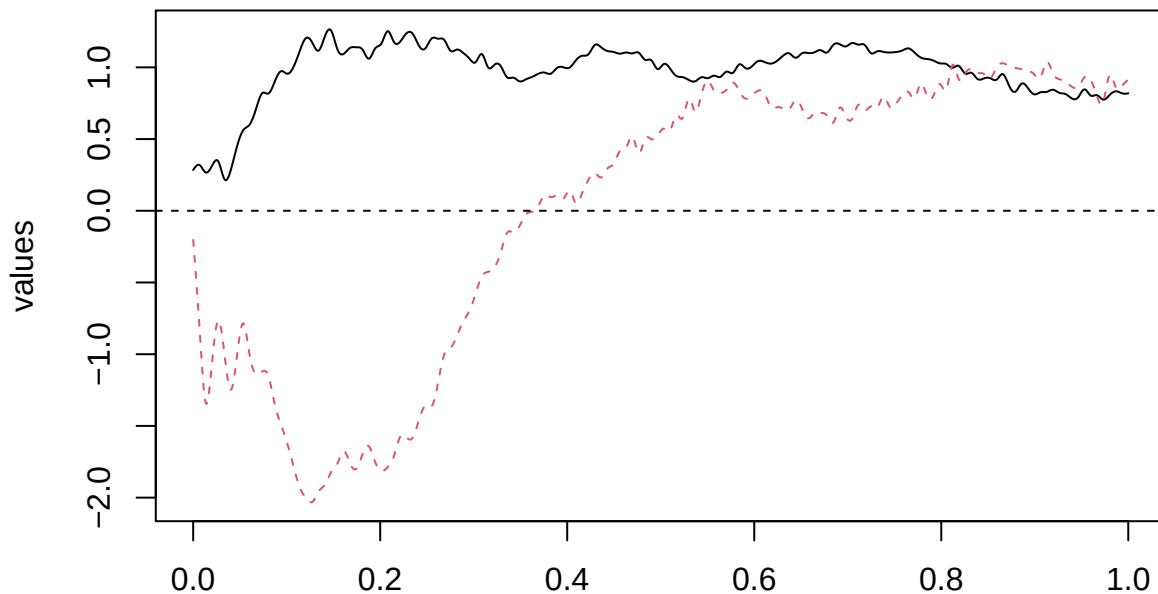
```
## 'argvals' is missing; using seq( 0 , 1 , length= 150 )
```

```
datos %>%
```

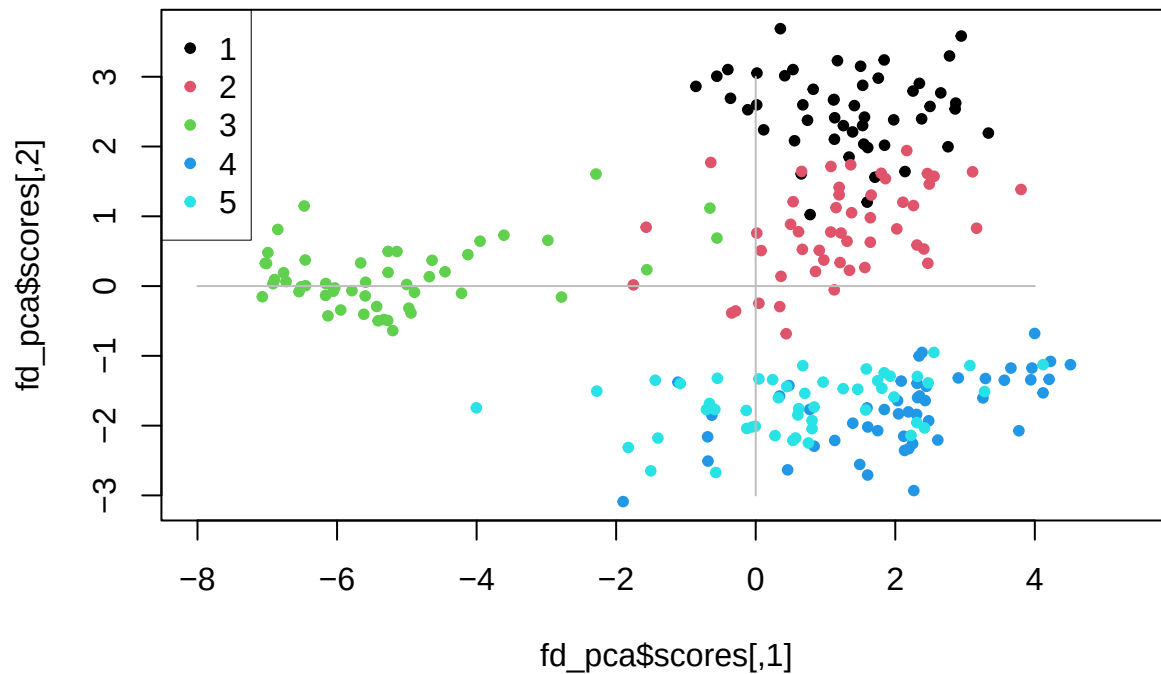
```
  ggplot(aes(variable, value, col = factor(id), group = id)) +
  geom_line(alpha = .6, show.legend = F) +
  # facet_wrap(~y) +
  theme_bw()
```



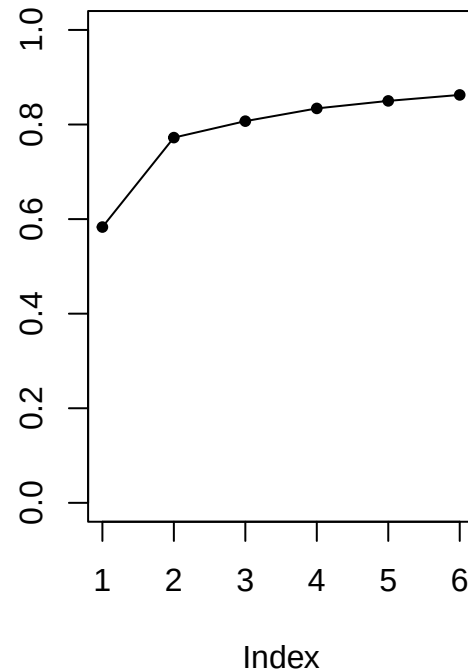
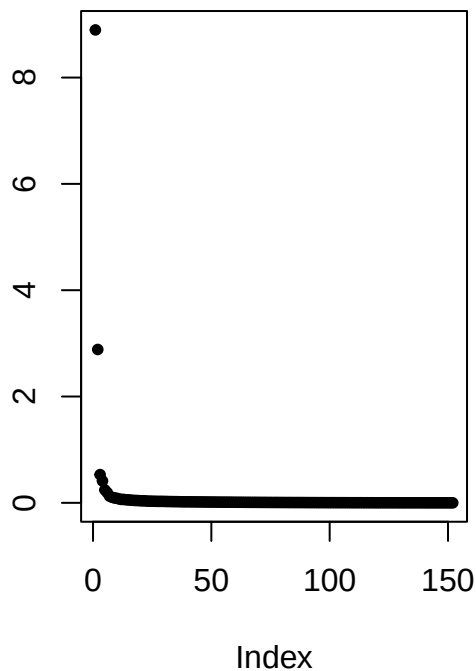
```
fd_pca <- pca.fd(fd, nharm = 6)
plot(fd_pca$harmonics[1:2]) %>% invisible
```



```
plot(fd_pca$scores, col = glearn, pch = 20, asp = 1, axes = T)
axis(1, pos = 0, labels = F, lwd.ticks = 0, col = 'grey')
axis(2, pos = 0, labels = F, lwd.ticks = 0, col = 'grey')
legend('topleft', legend = 1:5, col = 1:5, pch = 20, box.lwd = 0)
```



```
par(mfrow = c(1,2))
fd_pca$values %>% plot(pch = 20)
fd_pca$varprop %>% cumsum() %>% plot(pch = 20, ylim = c(0,1), type = 'o')
```



```
library(fdapace)

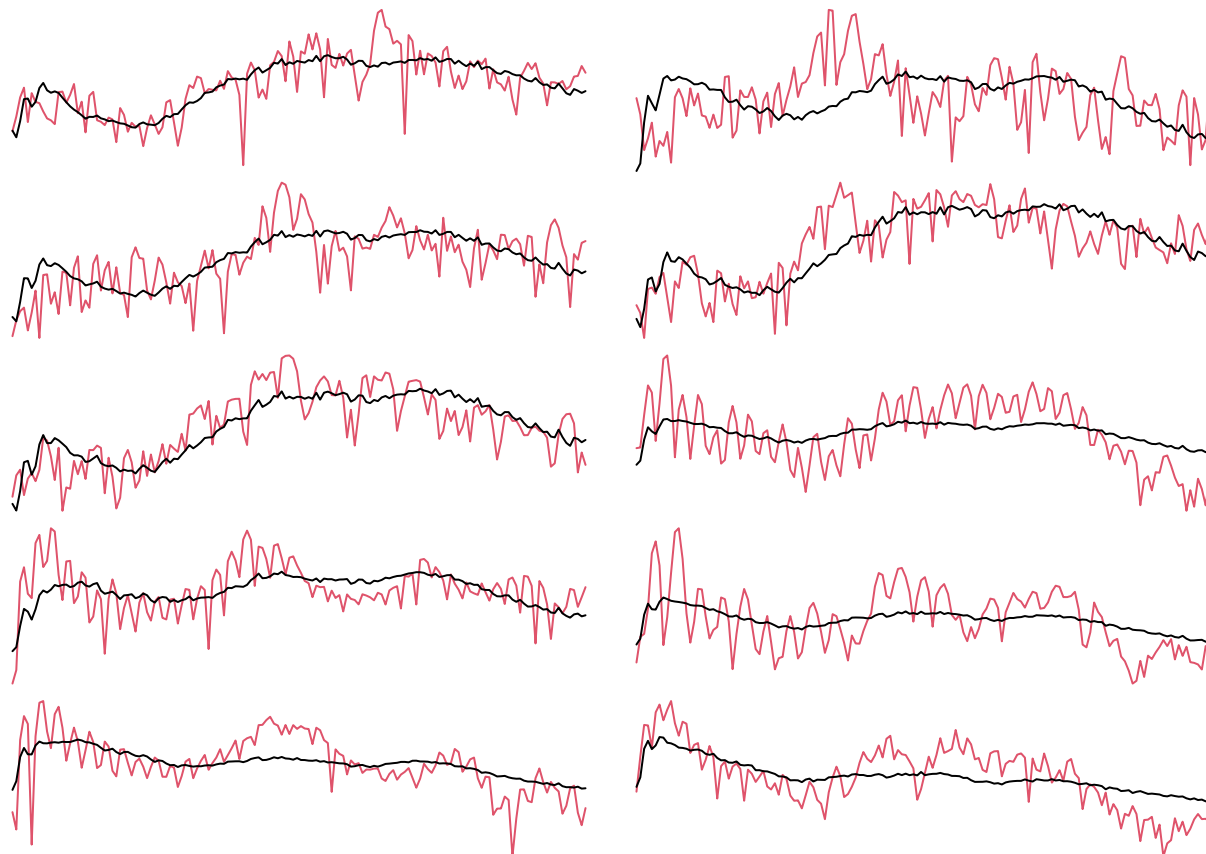
a <- Sparsify(mlearn$data, 1:150, 150)
fpca <- FPCA(a$Ly, a$Lt)

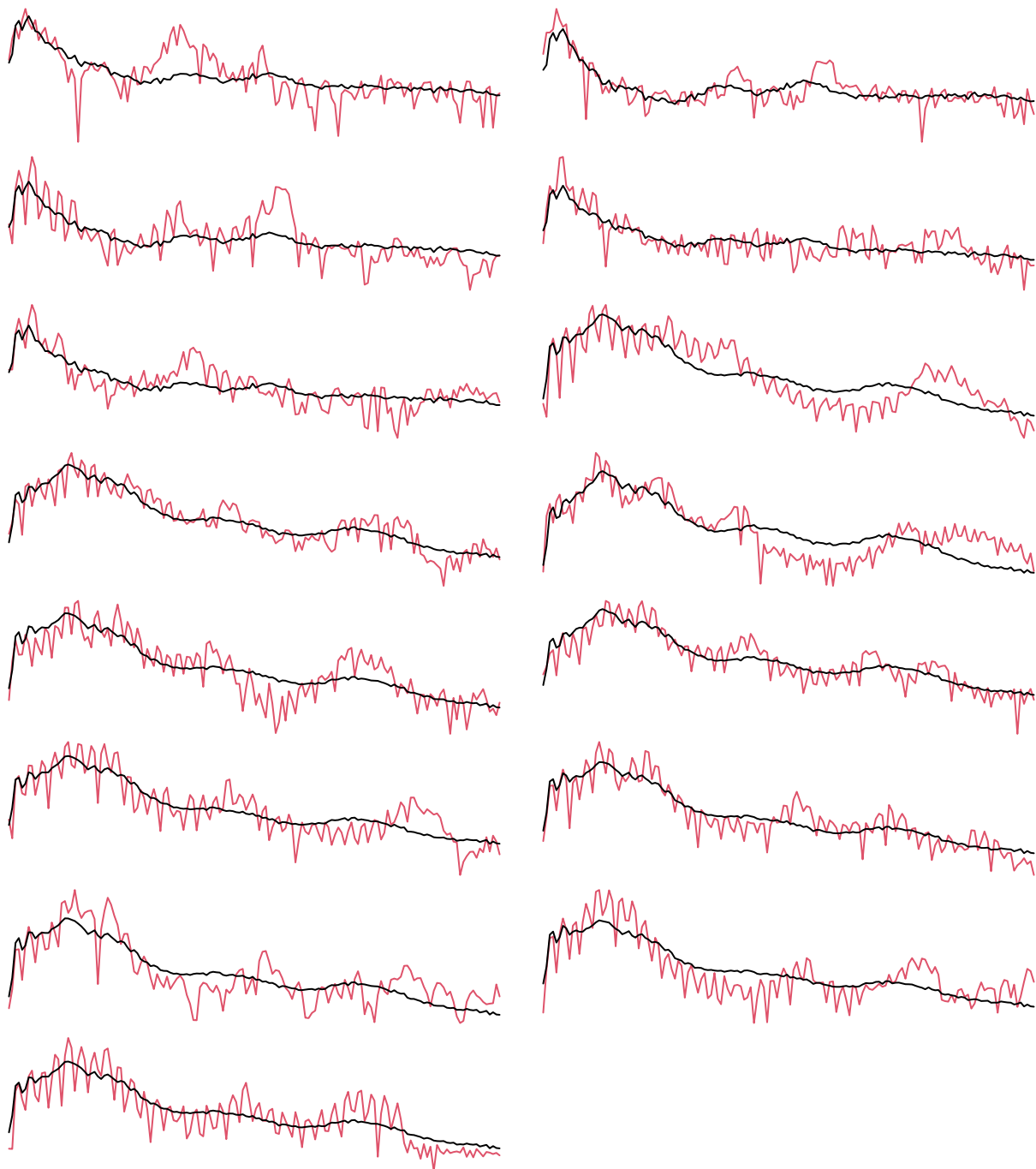
pred <- predict(fpca, newLt = a$Lt, newLy = a$Ly, K = 2)
```

```

# i <- 1
# plot(mlearn$data[i,], col = 2, type = 'l')
# lines(pred$predGrid, pred$predCurves[i,], type = 'l')
par(mfrow = c(5,2), mar = c(.1,.1,.1,.1))
sapply(seq(1,250,10), function(i){
  plot(mlearn$data[i,], col = 2, type = 'l', axes = F)
  lines(pred$predGrid, pred$predCurves[i,], type = 'l')
}) %>% invisible

```





```
pred2 <- predict(fpca, newLt = a$Lt, newLy = a$Ly, K = 10)
```

```
par(mfrow = c(5,2), mar = c(.1,.1,.1,.1))
sapply(seq(1,250,10), function(i){
  plot(mlearn$data[i,], col = 2, type = 'l', axes = F)
  lines(pred2$predGrid, pred2$predCurves[i,], type = 'l')
}) %>% invisible
```



